

SCIENTIST Born 1879 Died 1955

Albert Einstein

“The most **beautiful** experience we can have is the **mysterious**. It is the source of all true **art and science**.”

ALBERT EINSTEIN, 1930

German-born physicist Albert Einstein had a largely uninspiring school career, although he displayed a natural talent for mathematics and a love of learning. After graduating from a technical university with a teaching certificate, Einstein found a job at a patent office in Bern, Switzerland, in 1903. Two years later, while still working there, Einstein submitted his doctoral thesis and published four scientific papers. Three of these were to be profoundly important in the history of science.

The first of Einstein's 1905 papers concerned the puzzling phenomenon known as the photoelectric effect, in which electrons are ejected from atoms of metal when light shines onto them.

Instead of the ejection rate gradually building with the intensity of light shining onto the metal surface, the effect suddenly “switches on” when the frequency of light rises past a certain threshold. In order to explain this effect, Einstein showed that light energy must be delivered in discrete packets, or “quanta.”

German physicist Max Planck (1858–1947) had used quantization as a mathematical trick in 1900, but had

Einstein the icon

Albert Einstein contributed hugely to our understanding of the universe—from atoms to stars and galaxies—and became an iconic figure.



Young Einstein

Around the time of this photograph, Einstein—here with sister Maja—received a magnetic compass from his father. He later said that he had found inspiration wondering why compass needles turn.



School certificate

There is a myth that Einstein struggled at school. This certificate tells a different story—in physics and math he scored a maximum 6.

never supposed it to be literally true. Einstein's paper was the beginning of quantum theory—a field of physics that provides insight into the behavior of light, atoms, and subatomic particles. Einstein remained at the heart of its development for the rest of his life.

Einstein's second landmark paper of 1905, entitled *On the Electrodynamics of Moving Bodies*, outlined his Special

Theory of Relativity. The starting point in this paper was the fact that the speed of light is always the same, however fast an observer is moving relative to a light source. He used this to prove that time does not run at a single, “universal” rate—clocks moving relative to each other tick away seconds at different rates (though the effect is only noticeable at very high relative speeds).

Einstein's last publication of 1905 was a bold extension of Special Relativity. Working from the fact that the speed of light could not be exceeded, Einstein